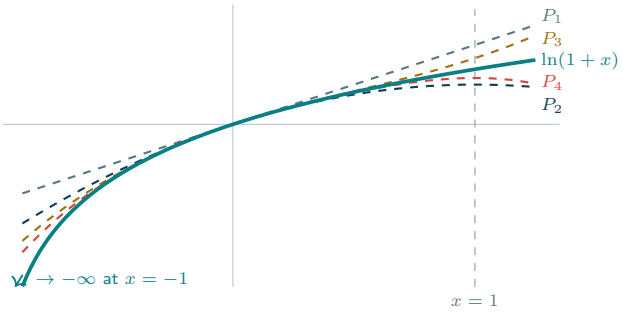
CALCULUS ON ONE PAGE

The logarithm as a power series

Integrating the geometric series once gives $\ln(1+x)$ and $\arctan x$, and integrating twice gives $\pi^2/12$ and Catalan's constant.

1 / THE RULE AND THE PICTURE

A function as a limit of polynomials, built by integrating a geometric series



partial sums $P_N(x)=\sum_{n\le N}\frac{(-1)^{n-1}x^n}{n}$ close in on $\ln(1+x)$ inside $-1<x\le 1$

A **power series** $c_0+c_1x+c_2x^2+\cdots$ is a limit of polynomials, its partial sums. For $|t|<1$ the geometric series gives $\frac{1}{1+t}=1-t+t^2-\cdots$, and since the logarithm is an integral and $\int_0^x t^{n-1}dt=\frac{x^n}{n}$, integrating term by term brings in the denominators n:

$$\ln(1+x)=x-\tfrac{x^2}{2}+\tfrac{x^3}{3}-\tfrac{x^4}{4}+\cdots$$

for $-1<x\le 1$. Each added term of P_N bends the polynomial closer to the curve.

2 / WHY THE ARCTANGENT, AND WHY ODD DENOMINATORS

Putting ix in the logarithm series splits it into a real part and the arctangent

Replacing t by t^2 in the geometric series gives a second alternating series $\frac{1}{1+t^2}=1-t^2+t^4-\cdots$ with only even powers, and integrating t^{2k} gives $\frac{x^{2k+1}}{2k+1}$, so its integral $\arctan x=\int_0^x\frac{dt}{1+t^2}$ has odd powers and odd denominators. It is the same series in disguise. In the logarithm series put ix for x: the powers i^n are real for even n and imaginary for odd n, so the even terms form one series, the odd terms another, and

$$\ln(1+ix)=\tfrac{1}{2}\ln(1+x^2)+i\arctan x$$

The arctangent is the imaginary part of the logarithm, and it keeps exactly the odd n. At $x=1$ this says $\ln(1+i)=\frac{1}{2}\ln 2+i\frac{\pi}{4}$, so the two order-one constants $\ln 2$ and $\frac{\pi}{4}$ are the real and imaginary parts of one logarithm.

3 / ORDER: HOW MANY TIMES THE GEOMETRIC SERIES IS INTEGRATED

Order 1 gives $\ln 2$ and $\frac{\pi}{4}$, order 2 gives $\frac{\pi^2}{12}$ and Catalan's constant G

order	all integers: logarithm family	odd integers: arctangent family	value at $x=1$
0	$\frac{1}{1+t}=\sum(-t)^k$	$\frac{1}{1+t^2}=\sum(-1)^kt^{2k}$	none
1	$\ln(1+x)=\sum\frac{(-1)^{n-1}x^n}{n}$	$\arctan x=\sum\frac{(-1)^nx^{2n+1}}{2n+1}$	$\ln 2=0.6931$ $\frac{\pi}{4}=0.7854$
2	$\int_0^x\frac{\ln(1+t)}{t}\,dt=\sum\frac{(-1)^{n-1}x^n}{n^2}$	$\int_0^x\frac{\arctan t}{t}\,dt=\sum\frac{(-1)^nx^{2n+1}}{(2n+1)^2}$	$\frac{\pi^2}{12}=0.8225$ $G=0.9160$
3	one more integration	one more integration	$\frac{3}{4}\zeta(3)=0.9015$ $\frac{\pi^3}{32}=0.9689$

At $x=1$ the series are the alternating sums $\sum\frac{(-1)^{n-1}}{n^s}$ and $\sum\frac{(-1)^n}{(2n+1)^s}$ with s the order. The logarithm family at order 2 is $1-\frac{1}{4}+\frac{1}{9}-\cdots=\int_0^1\frac{\ln(1+t)}{t}\,dt$, and making the even terms of $1+\frac{1}{4}+\frac{1}{9}+\cdots=\frac{\pi^2}{6}$ negative changes that sum by twice their total $\frac{1}{4}\cdot\frac{\pi^2}{6}$, which leaves $\frac{\pi^2}{6}\left(1-\frac{1}{2}\right)=\frac{\pi^2}{12}$. The arctangent family at order 2 is $G=\int_0^1\frac{\arctan x}{x}\,dx$, for which no expression in π is known. In general the logarithm family gives multiples of π^s at even orders and the arctangent family at odd orders, and the other orders produce new constants: $\ln 2$, G, $\zeta(3)$.

4 / ENDPOINT AND RADIUS

A finite identity controls the error at $x=1$, and the nearest complex singularity sets the radius

Integrating $\frac{1}{1+t}=\sum_{k<N}(-t)^k+\frac{(-t)^N}{1+t}$ gives $\ln(1+x)-P_N(x)=(-1)^N\int_0^x\frac{t^N}{1+t}\,dt$, so for $0\le x\le 1$ the error is at most $\frac{x^{N+1}}{N+1}$ and $\ln 2=1-\frac{1}{2}+\frac{1}{3}-\cdots$ with $|\ln 2-S_N|\le\frac{1}{N+1}$. The radius is 1 because $\frac{1}{1+t}$ has a pole at $t=-1$, and $\frac{1}{1+t^2}$ has poles at $t=\pm i$, also at distance 1, so $\arctan x$, smooth on the whole real line, has a series that fails beyond $|x|=1$.

5 / MAKING IT FAST

Subtracting the series for $\ln(1-x)$ keeps only the odd terms

Since $\ln\frac{1+x}{1-x}=2\left(x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots\right)$ and $x=\frac{1}{3}$ gives $\frac{1+x}{1-x}=2$, we get $\ln 2=2\left(\frac{1}{3}+\frac{1}{3\cdot 3^3}+\frac{1}{5\cdot 3^5}+\cdots\right)$, and five terms already give 0.693146, whereas the alternating series needs about 10^6 terms for six decimals.

ASIDE / A LOGARITHM IN A COUNTING SEQUENCE

The Catalan numbers $1,2,5,14,42,\ldots$ have reciprocals with $1-\frac{1}{2}+\frac{1}{5}-\frac{1}{14}+\cdots=\frac{11}{25}+\frac{24}{25\sqrt{5}}\ln\varphi\approx 0.6466$.

LINKS TO OTHER SHEETS

The Euler sheet treats $e^{i\theta}$ and the golden-metallic sheet the golden ratio φ.

Convention. P_N the Nth partial sum; order s counts integrations of $\frac{1}{1+t}$ or $\frac{1}{1+t^2}$; all values, identities and error bounds were checked numerically.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames